

Final Validation Report

Customer Personality Segmentation Project

Module 05 — Feature Engineering & Data Preprocessing (CPS-M05)

1. Objective

This report verifies that the final engineered dataset is fully ready for customer segmentation (clustering), confirming there are no missing values, unresolved duplicates, incorrect data types, or unscaled/unencoded features remaining.

2. Validation Checklist

Validation Check	Status	Notes
Missing Values	PASS	0 missing values across all 60 final columns.
Duplicate Records	PASS (reviewed)	183 rows share an identical feature profile after the unique ID column was removed. Investigation confirmed these are genuine repeated customer profiles (e.g. shared median-imputed Income and matching zero/low spending-and-purchase activity), not pipeline errors — intentionally kept so cluster sizes reflect the true customer count.
Data Types	PASS	All 60 columns are numeric (int64/float64); no object/text columns remain.
Feature Consistency	PASS	All 2,237 rows share the same 60 consistent columns; no ragged or missing feature groups.
Scaling Completion	PASS	All continuous features scaled with RobustScaler; medians centered near zero.
Encoding Completion	PASS	All categorical variables converted to numeric via label or one-hot encoding.

3. Final Dataset Summary

Property	Value
Input (cleaned) columns	33
Engineered features added	13
Encoded (one-hot/label) columns added	12
Columns removed (identifiers/redundant/constant)	16
Log-transformed variant columns added	11
Final column count	60
Final row count	2,237
Scaling method	RobustScaler

4. Scaler Selection Justification

RobustScaler was selected over StandardScaler and MinMaxScaler because it centers features on the median and scales by the interquartile range (IQR), making it far less sensitive to the small number of remaining high-value outliers (e.g. very high spenders) than mean/standard-deviation or min/max-based scaling. This produces more stable, comparable feature ranges for distance-based clustering algorithms such as K-Means.

5. Conclusion

The dataset `customer_ml_ready_dataset.csv` (2,237 rows $\times$ 60 columns) has passed all validation checks and is ready to be used directly as input for clustering-based customer segmentation.